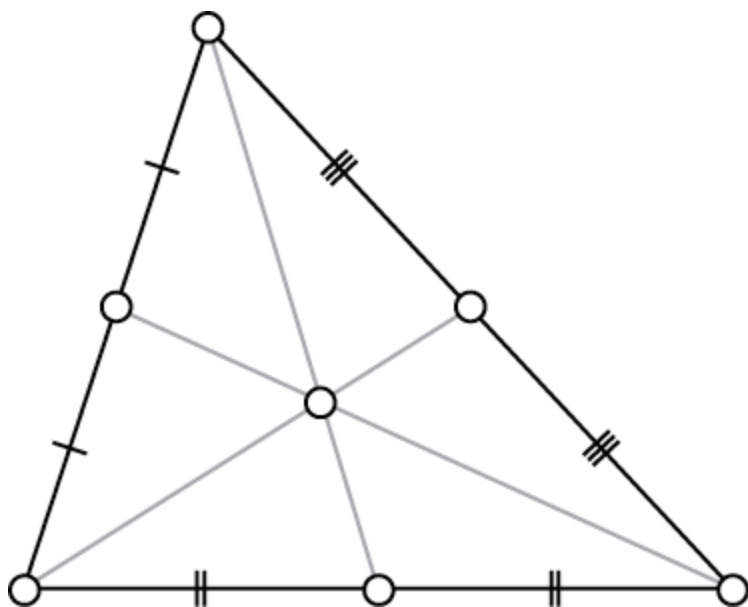




[Figure 7-5](#): Triangle centroid (source Wikimedia Commons)

Remarkably, all the lines you drew meet in a single point that looks something like the “center” of the triangle. All three lines will meet at a single point no matter what triangle you start with. The point where they meet is commonly called the *centroid* of the triangle, and it’s always on the inside in a place that looks like it could be called the triangle’s center.

Some shapes, like circles, always have one point that can unambiguously be called the shape’s center. But triangles aren’t like this: the centroid is one center-ish point, but there are other points that could also be considered centers. Consider this new algorithm for any triangle:

1. Bisect each side of the triangle.
2. Draw a line perpendicular to each side through the side’s midpoint.

In this case, the lines do not typically go through the vertices like they did when we drew a centroid. Compare [Figure 7-5](#) with [Figure 7-6](#).